class12

Joli DeLucia A17320476

Q.13

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271	NA19144	G/G	24.85165
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279	NA20508	A/G	21.29782
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326	HG00171	A/G	21.09331
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460 HG00119      A/G   31.53069
461 NA19130      A/A   44.27738
462 HG00239      A/G   23.18250
```

```r
geno_counts <- table(dat$genotype)
geno_counts
```

```
A/A A/G G/G 
108 233 121 
```

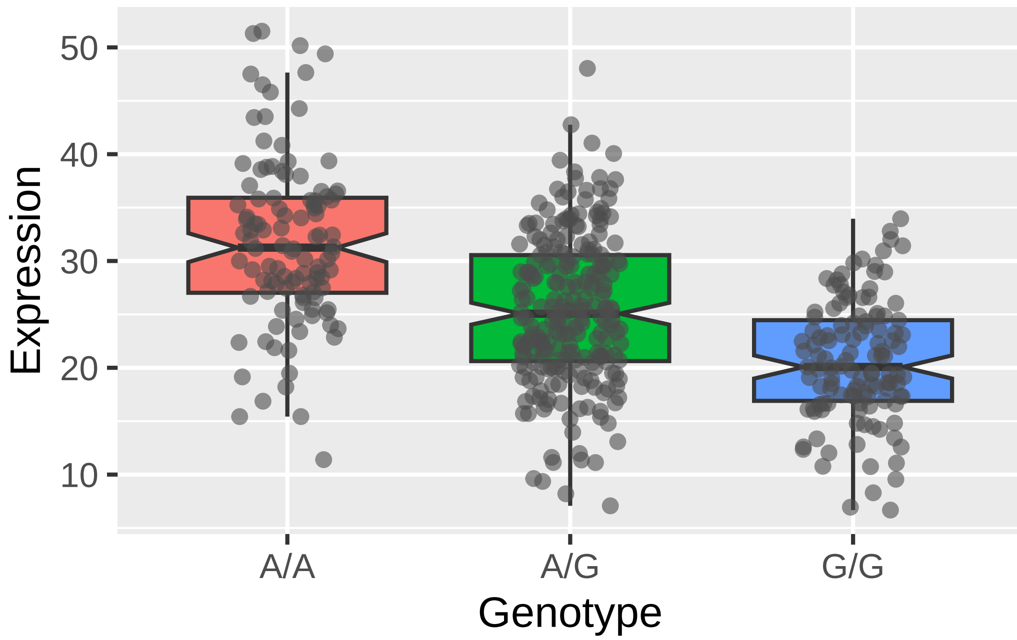
```r
geno_medians <- tapply(dat$expression, dat$genotype, median)
geno_medians
```

```
     A/A      A/G      G/G 
31.24847 25.06486 20.07363 
```

Q.14

```r
library(ggplot2)
dat$genotype <- factor(dat$genotype, levels = c("A/A","A/G","G/G"))

ggplot(dat, aes(x = genotype, y = expression, fill = genotype)) +
  geom_boxplot(notch = TRUE, width = 0.70, outlier.shape = NA) +
  geom_jitter(width = 0.18, alpha = 0.6, size = 2, color = "gray30") +
  scale_fill_manual(values = c("A/A"="#F8766D", "A/G"="#00BA38", "G/G"="#619CFF")) +
  labs(x = "Genotype", y = "Expression") +
  theme_gray(base_size = 16) +
  theme(legend.position = "none")
```


50
40
30
20
10
Expression
A/A
A/G
G/G
Genotype